EXP 5

Aim:

To design and simulate a microwave impedance-matching network for a complex load of $75+j25\,\Omega$ matched to a $50\,\Omega$ reference impedance at $10\,\text{GHz}$, and to design a $50\,\Omega$ frequency-selective filter with a cutoff frequency of $10\,\text{GHz}$. To evaluate and analyze the return loss, insertion loss, S_{11}, S_{21}, passband, stopband, and bandwidth over the frequency range of $8\text{–}12\,\text{GHz}$ using MATLAB.

Software required:

Matlab

Program code:

```
%% EXP.5: MICROWAVE IMPEDANCE MATCHING AND FILTER DESIGN
clc;
clear;
close all;

set(0,'DefaultFigureVisible','on');

%% 1. INPUT PARAMETERS FROM EXP.5 WORD DOCUMENT

f0 = 10e9;                    % Design frequency: 10 GHz
Z0 = 50;                      % Reference impedance: 50 ohms
ZL = 75 + 1j*25;              % Load impedance: 75+j25 ohms
fc = 10e9;                    % Filter cutoff: 10 GHz
fstart = 8e9;                 % Sweep start: 8 GHz
fstop = 12e9;                 % Sweep stop: 12 GHz
N = 5;                        % Butterworth filter order
ylim([-60 5]);
drawnow;


%% 7. FIGURE 3: INPUT IMPEDANCE AFTER MATCHING


figure(3);
clf;
set(gcf,'Name','Input Impedance');


subplot(2,1,1);
plot(f/1e9,real(ZinAfter),'LineWidth',2);
hold on;
plot([8 12],[50 50],'r--','LineWidth',1.2);
grid on;
xlabel('Frequency (GHz)');
ylabel('Real Z_{in} (ohms)');
title('Real Part of Input Impedance');
legend('Real Z_{in}','50 ohms','Location','best');


subplot(2,1,2);
plot(f/1e9,imag(ZinAfter),'LineWidth',2);
hold on;
```

```
c = 3e8;                        % Speed of
light


f = linspace(fstart,fstop,1001);
lambda0 = c/f0;
beta0 = 2*pi/lambda0;


%% 2. FIGURE 1: OBJECTIVES DIAGRAM


figure(1);
clf;
set(gcf,'Name','EXP.5 Objectives');


axis([0 10 0 10]);
axis off;
hold on;


text(5,9.3,'EXP.5: OBJECTIVES',...

'FontSize',16,'FontWeight','bold',...
    'HorizontalAlignment','center');


objectives = {
    '1. Design an impedance-matching
network'
    '2. Minimize reflection and
maximize power transfer'
    '3. Design a 50-ohm microwave
filter'
    '4. Evaluate return loss,
insertion loss and bandwidth'
    };


ypos = [7.5 5.8 4.1 2.4];


for k = 1:4
    rectangle('Position',[0.8 ypos(k)-
0.4 8.4 0.9],...
        'FaceColor',[0.85 0.92 1],...

'EdgeColor','b','LineWidth',1.5);
plot([8 12],[0 0],'r--
','LineWidth',1.2);
grid on;
xlabel('Frequency (GHz)');
ylabel('Imaginary Z_{in} (ohms)');
title('Imaginary Part of Input
Impedance');
legend('Imaginary Z_{in}','Zero
reactance','Location','best');


drawnow;


%% 8. FILTER DESIGN: IDEAL BUTTERWORTH
LOW-PASS MODEL


% Ideal theoretical Butterworth
response.
% Not a physical circuit or
electromagnetic simulation.


S21power = 1./(1+(f/fc).^(2*N));


S21mag = sqrt(S21power);
S11mag = sqrt(1-S21power);


% S-parameter magnitudes in dB
S21dB = 20*log10(max(S21mag,1e-12));
S11dB = 20*log10(max(S11mag,1e-12));


% Return loss is positive: RL = -
20*log10(abs(S11))
FilterReturnLoss = -S11dB;


% Insertion loss is positive: IL = -
20*log10(abs(S21))
FilterInsertionLoss = -S21dB;


% Ideal Butterworth -3 dB cutoff
fcSim = fc;
deviation = abs(fcSim-fc)/fc*100;
```

```
    text(5,ypos(k),objectives{k},...
'HorizontalAlignment','center','FontSi
ze',10);
end


text(5,0.8,...
    'ZL = 75+j25 ohms | Z0 = 50 ohms |
fc = 10 GHz',...
'HorizontalAlignment','center','FontSi
ze',9);


drawnow;


%% 3. IMPEDANCE MATCHING BEFORE
MATCHING


GammaL = (ZL-Z0)/(ZL+Z0);
GammaMagBefore = abs(GammaL);


% Return loss is positive for a
passive mismatched load
RLbefore = -
20*log10(max(GammaMagBefore,1e-12));


fprintf('\n==========================
============\n');
fprintf(' PART A: IMPEDANCE
MATCHING\n');
fprintf('==========================
==========\n');


fprintf('Design frequency       = %.2f
GHz\n',f0/1e9);
fprintf('Reference impedance    = %.2f
ohms\n',Z0);
fprintf('Load impedance         = %.2f
%+.2fj ohms\n',...
    real(ZL),imag(ZL));
```

```
%% 9. FIGURE 4: FILTER S11 AND S21


figure(4);
clf;
set(gcf,'Name','Filter S Parameters');


plot(f/1e9,S11dB,'LineWidth',2);
hold on;
plot(f/1e9,S21dB,'LineWidth',2);


plot([8 12],[-3.0103 -3.0103],'r--
','LineWidth',1.2);
plot([10 10],[-60 5],'k--
','LineWidth',1.2);


grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Filter S11 and S21 Frequency
Response');
legend('S11 (dB)','S21 (dB)','-3 dB
level',...
    'Cutoff 10
GHz','Location','best');
xlim([8 12]);
ylim([-60 5]);
drawnow;


%% 10. FIGURE 5: RETURN LOSS


figure(5);
clf;
set(gcf,'Name','Return Loss');


plot(f/1e9,RLafter,'LineWidth',2);
hold on;
plot(f/1e9,FilterReturnLoss,'LineWidth
',2);
```

```
fprintf('Reflection coefficient =
%.6f\n',GammaMagBefore);
fprintf('Return loss before matching =
%.4f dB\n',RLbefore);


%% 4. SINGLE SHUNT SHORT-CIRCUITED
STUB DESIGN


yL = Z0/ZL;


dvec = linspace(1e-8,lambda0/2-1e-
8,20001);


ylineLoad = (yL + 1j*tan(beta0*dvec))
./ ...
    (1 + 1j*yL.*tan(beta0*dvec));


err = real(ylineLoad)-1;
idx = find(err(1:end-
1).*err(2:end)<=0,1);


if isempty(idx)
    error('Matching distance not
found.');
end


d = fzero(@(x) real( ...
    (yL+1j*tan(beta0*x))/ ...
    (1+1j*yL*tan(beta0*x)))-1,...
    [dvec(idx),dvec(idx+1)]);


yd = (yL+1j*tan(beta0*d))/ ...
    (1+1j*yL*tan(beta0*d));


b = -imag(yd);


% Short-circuited shunt stub
theta = atan2(1,-b);
l = theta/beta0;
```

```
plot([10 10],[0 60],'k--
','LineWidth',1.2);


grid on;
xlabel('Frequency (GHz)');
ylabel('Return Loss (dB)');
title('Return Loss of Matching Network
and Filter');
legend('Matching network after
matching',...
    'Filter return loss from S11',...
    '10 GHz','Location','best');
xlim([8 12]);
drawnow;


%% 11. FIGURE 6: FILTER INSERTION LOSS


figure(6);
clf;
set(gcf,'Name','Insertion Loss');


plot(f/1e9,FilterInsertionLoss,'LineWi
dth',2);
hold on;


plot([8 12],[2 2],'r--
','LineWidth',1.2);
plot([10 10],[0
max(FilterInsertionLoss)+1],...
    'k--','LineWidth',1.2);


grid on;
xlabel('Frequency (GHz)');
ylabel('Insertion Loss (dB)');
title('Filter Insertion Loss versus
Frequency');
legend('Insertion loss','2 dB
reference',...
    'Cutoff = 10
GHz','Location','best');
xlim([8 12]);
drawnow;
```

```
fprintf('\n===== STUB DESIGN =====\n');
fprintf('Distance from load d = %.4f mm\n',d*1000);
fprintf('Stub length l        = %.4f mm\n',l*1000);


%% 5. MATCHING NETWORK FREQUENCY RESPONSE


GammaBefore = zeros(size(f));
GammaAfter = zeros(size(f));
ZinAfter = zeros(size(f));


for k = 1:length(f)


    beta = 2*pi*f(k)/c;


    GammaBefore(k) = (ZL-Z0)/(ZL+Z0);


    yd_f = (yL+1j*tan(beta*d))/ ...
        (1+1j*yL*tan(beta*d));


    ystub = -1j*cot(beta*l);
    ytotal = yd_f+ystub;


    GammaAfter(k) = (1-ytotal)/(1+ytotal);
    ZinAfter(k) = Z0/ytotal;
end


% Return loss after matching, in positive dB
RLafter = -20*log10(max(abs(GammaAfter),1e-12));


[~,i0] = min(abs(f-f0));
```

```
%% 12. OUTPUT VALUES AT 8, 10 AND 12 GHz


testFreq = [8 10 12]*1e9;


fprintf('\n============================================\n');
fprintf(' PART B: FILTER RESULTS\n');
fprintf('============================================\n');


fprintf('Filter impedance = %.2f ohms\n',Z0);
fprintf('Butterworth order = %d\n',N);
fprintf('Specified cutoff frequency = %.2f GHz\n',fc/1e9);
fprintf('Calculated ideal cutoff frequency = %.2f GHz\n',...
    fcSim/1e9);
fprintf('Cutoff percentage deviation = %.4f %%\n',deviation);


fprintf('\nFrequency | Return Loss | Insertion Loss\n');
fprintf('  (GHz)   |    (dB)    |    (dB)\n');
fprintf('-----------------------------------------\n');


for k = 1:length(testFreq)
    [~,ii] = min(abs(f-testFreq(k)));


    fprintf('  %5.2f   | %9.4f  | %10.4f\n',...
        f(ii)/1e9,...
        FilterReturnLoss(ii),...
        FilterInsertionLoss(ii));
end


%% 13. FINAL OUTPUT SUMMARY
```

```
fprintf('\n===== MATCHING RESULTS AT 10 GHz =====\n');
fprintf('Input impedance after matching = %.4f %+.4fj ohms\n',...
real(ZinAfter(i0)),imag(ZinAfter(i0)));
fprintf('Return loss before matching = %.4f dB\n',RLbefore);
fprintf('Return loss after matching  = %.4f dB\n',RLafter(i0));
fprintf('Reflection after matching   = %.6f\n',...
    abs(GammaAfter(i0)));

%% 6. FIGURE 2: REFLECTION RESPONSE

figure(2);
clf;
set(gcf,'Name','Impedance Matching Response');

plot(f/1e9,20*log10(max(abs(GammaBefore),1e-12)),...
    'LineWidth',2);
hold on;

plot(f/1e9,20*log10(max(abs(GammaAfter),1e-12)),...
    'LineWidth',2);

plot([8 12],[-15 -15],'r--','LineWidth',1.2);
plot([10 10],[-60 5],'k--','LineWidth',1.2);

grid on;
xlabel('Frequency (GHz)');
ylabel('Reflection coefficient magnitude (dB)');
```

```
fprintf('\n============================================\n');
fprintf(' EXP.5 FINAL OUTPUT SUMMARY\n');
fprintf('============================================\n');

fprintf('Load reflection coefficient = %.5f\n',GammaMagBefore);
fprintf('Return loss before matching = %.4f dB\n',RLbefore);
fprintf('Return loss after matching at 10 GHz = %.4f dB\n',...
    RLafter(i0));

fprintf('Input impedance after matching = %.4f %+.4fj ohms\n',...
real(ZinAfter(i0)),imag(ZinAfter(i0)));

fprintf('Filter return loss at 10 GHz = %.4f dB\n',...
    FilterReturnLoss(i0));

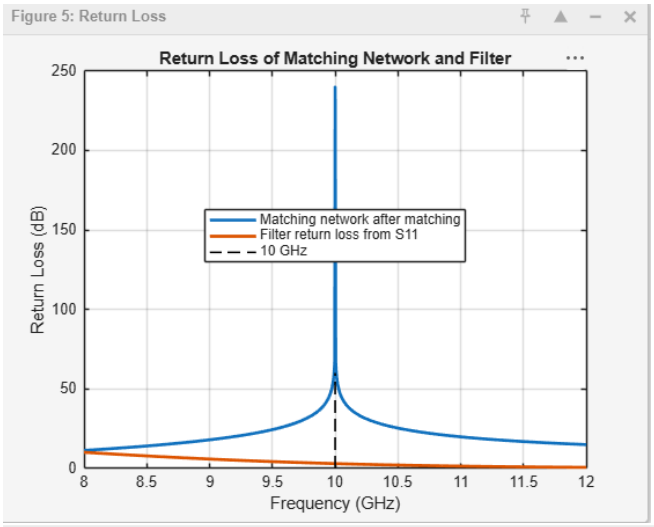
fprintf('Filter insertion loss at 10 GHz = %.4f dB\n',...
    FilterInsertionLoss(i0));

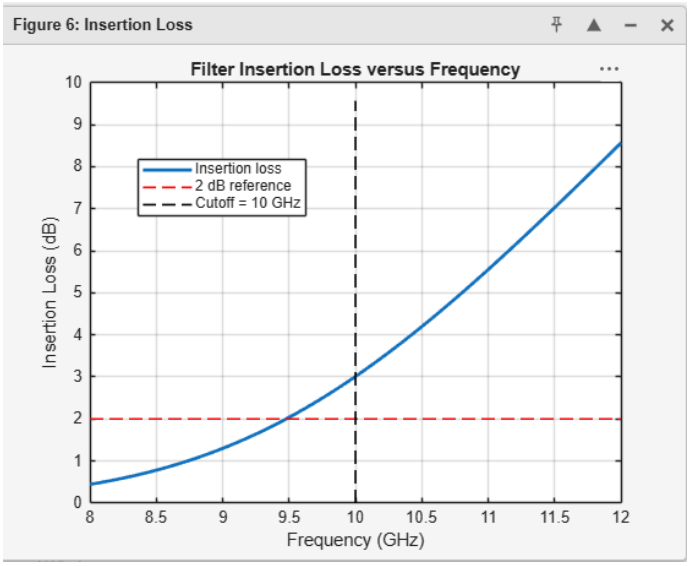
fprintf('Filter cutoff frequency = %.4f GHz\n',fcSim/1e9);
fprintf('Cutoff deviation = %.4f %%\n',deviation);

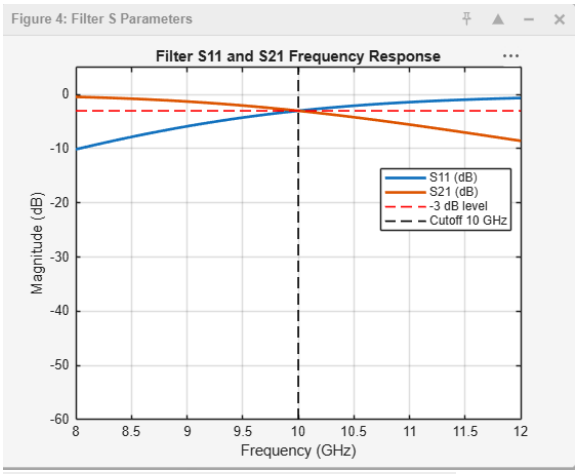
fprintf('\nAll six figures generated.\n');
```

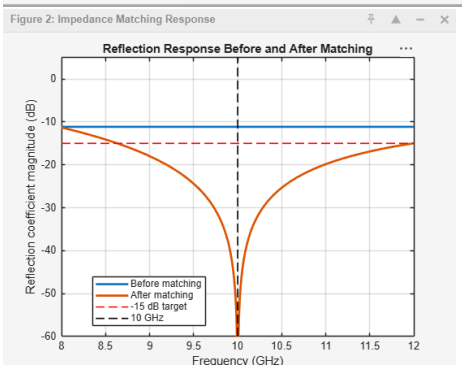
```
title('Reflection Response Before and
After Matching');
legend('Before matching','After
matching',...
    '-15 dB target','10
GHz','Location','best');
xlim([8 12]);
```

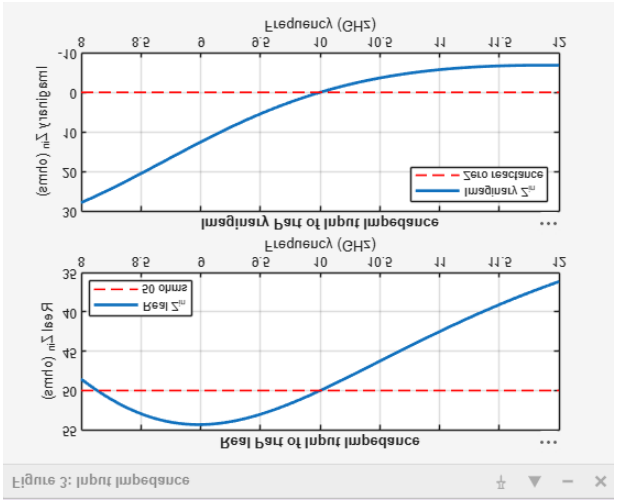
Output:






Figure 4: Filter S Parameters
Filter S11 and S21 Frequency Response
S11 (dB)
S21 (dB)
-3 dB level
Cutoff 10 GHz
Magnitude (dB)
Frequency (GHz)


Figure 2: Impedance Matching Response
Reflection Response Before and After Matching
Before matching
After matching
-15 dB target
10 GHz
Reflection coefficient magnitude (dB)
Frequency (GHz)


Figure 3: Input Impedance
Imaginary Part of Input Impedance
Zero reactance
Imaginary Z_in
Imaginary Z_in (ohms)
Frequency (GHz)
Real Part of Input Impedance
20 ohms
Real Z_in
Real Z_in (ohms)
Frequency (GHz)